

SHARANYA MISHRA

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Portfolio: <https://www.sharanya-mishra.com/>

EDUCATION

Illinois Institute of Technology, Chicago, IL	May 2026
<i>Master of Science- Data Science - GPA: 3.7</i>	
PES University, Bengaluru, India	June 2024
<i>Bachelor in technology - Computer Science/Engineering-GPA: 3.5</i>	

SKILLS

Programming & Tools: Python, R, SQL, C++, JavaScript, MATLAB, Git, Docker, Kubernetes
AI / ML & Data Science: Machine Learning, Deep Learning, NLP, Computer Vision, Regression, Time-Series Forecasting (ARIMA, SARIMA, GARCH), Feature Engineering, Model Evaluation
Data Analysis & Statistics: Probability, Hypothesis Testing, Statistical Inference, Process Optimization, Work Measurement, A/B Testing
Visualization & Reporting: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Excel (Pivot Tables, VLOOKUP, Variance Analysis), SAP
Cloud & Big Data: AWS (S3, EC2, SageMaker), Microsoft Azure, PySpark, Hadoop, Jenkins

EXPERIENCE

Lead Research Assistant <i>Illinois Institute of Technology</i>	January 2025 - Present
- Led a 4-member team in developing OptiWeb, a browser-based trajectory optimization platform for robotics applications. - Automated optimization model generation and benchmarked performance across 50+ NEOS solver configurations. - Improved simulation performance by ~30% and integrated LLM-driven assistance for optimization setup and analysis.	

Data Science Consultant / Project Lead LabelMaster Practicum	January 2026 – May 2026
- Analyzed multi-year customer, transaction, product, and location datasets (2012–2026) to study churn, retention, and revenue concentration. - Identified a key trend: active customers declined while revenue per customer increased, highlighting account concentration risk. - Built customer segmentation tiers using revenue contribution, purchase activity, geography, and dormant-account behavior. - Found win-back opportunities among inactive but historically valuable customers across priority states. - Presented stakeholder-ready recommendations for retention campaigns, geographic targeting, and customer reactivation strategy.	

Data Classification- Spring Intern <i>Aptos Retail</i>	January 2024 - July 2024
- Built 15+ Azure-based data validation rules to classify sensitive enterprise data and improve reporting accuracy to 95%. - Automated classification workflows across 5+ teams using Microsoft Purview and Salesforce, reducing manual effort by 40. - Supported data governance by improving data visibility, ownership tracking, and compliance-ready reporting.	

Data Analyst Summer Intern (Risk Advisory) <i>Deloitte Touche Tohmatsu LLC</i>	June 2023 - August 2023
Automated anomaly detection across 7+ client datasets using Python and SQL to identify reporting inconsistencies. - Built Excel dashboards using Pivot Tables, VLOOKUP, and variance analysis to support risk advisory decisions. - Cleaned and validated client data to improve audit-readiness and reporting efficiency.	

Data Analyst Intern <i>Capgemini</i>	June 2022 - August 2022
Developed SQL-based analytics pipelines and Tableau dashboards, enhancing client reporting efficiency.	

RESEARCH PUBLICATION

Audio Description for the Visually Impaired (IEEE, 2024)
Link to paper: <https://ieeexplore.ieee.org/document/10544216>
GitHub: https://github.com/Obsarian/Audio-Description-of-Videos_Capstone
Technologies: Python, PyTorch, YOLO, ResNet-35 (Kinetics), Google Text-to-Speech API

- Developed a **deep learning pipeline** to generate **real-time voice captions** from videos, achieving **BLEU score 0.7**.
- Combined **object and action recognition** models to enhance **video accessibility** for visually impaired users.

PROJECT

HerWay - AI-Powered Urban Analytics Platform GitHub: github.com/mishrasharanya/HerWay-Soremo Live Demo: https://her-way-soremo.vercel.app/ Technologies: Python, NLP, OpenAI API, React, Leaflet	
- Built a multi-source urban analytics platform combining Reddit, 150k+ 311 service requests, and crime datasets to generate neighborhood-level insights across 77 Chicago community areas - Developed one of the only fellowship projects integrating NLP and OpenAI-powered analysis to extract community signals from 1,000+ public posts and records. - Built an AI-powered chatbot and geospatial dashboard for community-level insights and decision making.	
ForeQuest - AI-Powered Financial Forecasting Platform GitHub: github.com/mishrasharanya/Forequest-Forecast_Smarter Live: https://forequest.streamlit.app/ Technologies: Python, Streamlit, GARCH, Monte Carlo, Groq, LLMs	
- Built an end-to-end financial forecasting platform integrating ARIMA, GARCH, Monte Carlo simulation, option pricing models, and 10+ years of market data for stock trend and volatility analysis. - Achieved ~85% R² and <0.02 MAE in volatility prediction while integrating Groq-hosted LLMs to deliver explainable forecast interpretation, risk analysis, and conversational financial insights through a deployed Streamlit application.	
OptiWeb - Browser-Based Trajectory Optimization Platform Research Project Midwest Robotics Workshop 2026 Technologies: React, Node.js, AMPL, JavaScript	
- Developed OptiWeb, a browser-based trajectory optimization platform for automated model generation and reproducible robotics experimentation. - Implemented cart-pole, five-link biped, and quadruped benchmarks while evaluating convergence across 50+ NEOS solver configurations. - Established quality assurance procedures to validate solution quality, convergence behavior, and reproducibility across benchmark robotic systems.	